

Why Simple Agentic RAG Wins

Today the model decides its own filters. One loop, two tools, answers still cited to the page. The elaborate agents cost extra calls on every question, so they wait for a failure you measured.

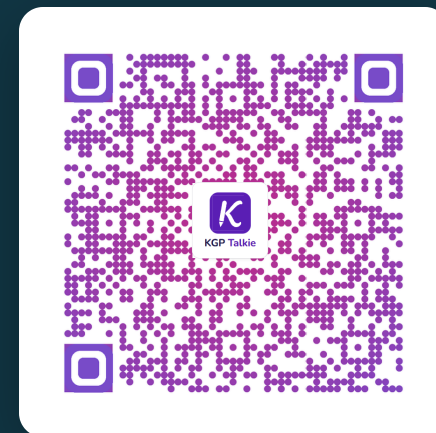
one loop

two tools

the model picks filters

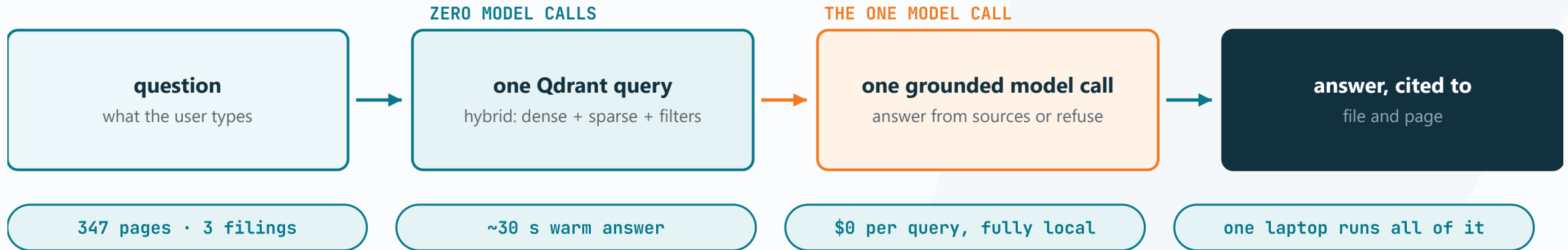
2 to 3 calls

cited to the page



Scan to subscribe

Simple RAG: the Baseline We **Already Have**



PAGE CHUNKS

One chunk is one page

So a citation is a page a reader can open and check.

HYBRID + FILTERS

Sparse, dense, filters

Sparse catches AWS and 10-K, dense catches meaning, filters narrow.

GROUNDED PROMPT

Rule 3 refuses

When sources fall short. The guardrail rides the one call, free of charge.

Good pages in, grounded answer out: retrieval is fixed **before** generation, so nothing needs correcting after.

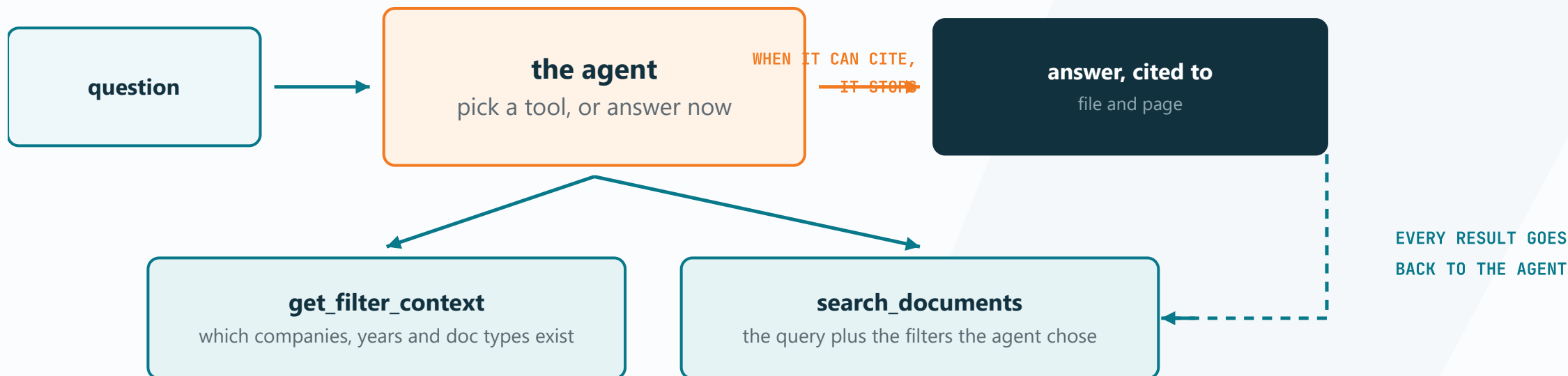
Where One Retrieval **Runs Out**

THE QUESTION A USER TYPES	WHAT ONE RETRIEVAL DOES	WHY THE ANSWER SLIPS
How much did Amazon spend on research and development in 2024?	one vector search across all three filings at once	Google R&D pages score just as high as Amazon
Compare AWS operating income with Google Cloud for 2024.	one query, one ranked list of five pages	two companies need two searches, not one
Which companies and years does this collection actually hold?	searches the text inside the filings	the answer lives in the metadata, not the text

The fix is not a longer prompt. Let the model retrieve more than once, and let it choose the filters. That is the whole idea of an agent.

All three failures happen **before generation**, in retrieval. A refusal is honest, but it is still not the answer the user wanted.

Agentic RAG: the Model Decides the **Filters**



What changed: auto_filter is off, so the filter dict is now the model's decision. Retrieval and the grounded prompt are the same code as before.

The loop is the only new machinery: the model reads the real stored values, then searches with filters it chose. **No router, no grader, no critique pass.**

Simple Agent Beats **Elaborate** Agent

This pipeline: the simple agent

- ✓ **Components:** 1 prompt, 2 tools
- ✓ **Model calls per question:** 2 to 3
- ✓ **Wait on a local 27B:** about 60 to 90 s
- ✓ **When the answer is wrong:** read 3 streamed steps
- ✓ You can hold the whole run in your head.

The tutorial architecture: the elaborate agent

- ✓ **Components:** planner, router, grader, critique loop, reranker, supervisor
- ✓ **Model calls per question:** 8 to 15
- ✓ **Wait on a local 27B:** 4 minutes and up
- ✓ **When the answer is wrong:** 6 places to look
- ✓ Six prompts, and every one of them can drift.

Both answer the same question. Only one of them can be **debugged at nine in the evening**.

The RAG Family, in One Look

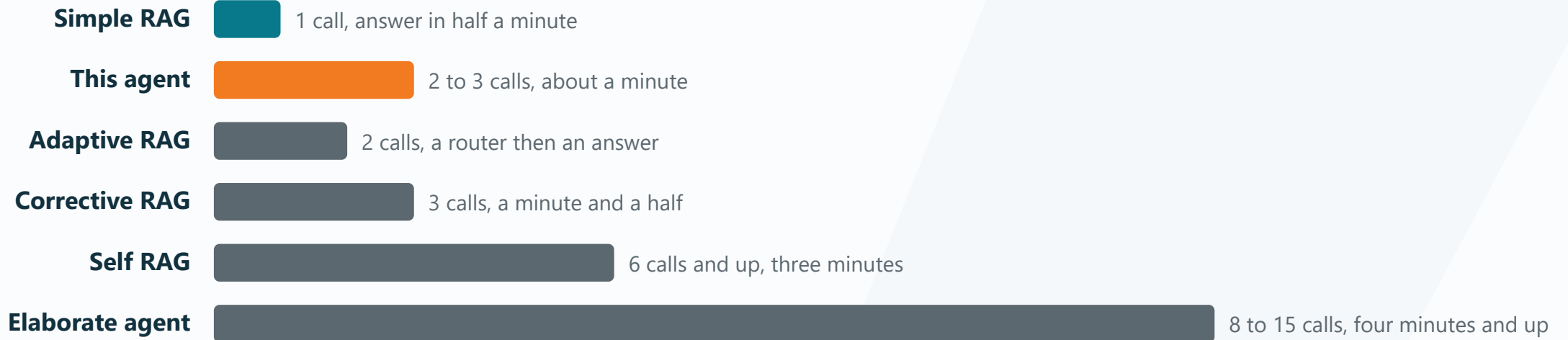


Count the orange boxes. The middle three fix a retrieval mistake **after** it happens. Our agent fixes retrieval **before** it happens, by choosing the filters.

What Extra Calls Cost

07

MODEL CALLS PER QUESTION • ONE CALL RUNS ABOUT 30 SECONDS ON A LOCAL 27B



LATENCY

Users feel every call

Half a minute survives a chat UI. Four minutes does not.

COST

Tokens are billed per call

A critique pass re-reads the same context. Six calls, six payments.

SCALABILITY

A GPU serves so many calls

Six calls per question serves one sixth of the users.

When the Complex Ones Earn Their Calls

Start simple: one retrieval, one model call. Measure first, then buy one upgrade with the failure you measured.

THE SYMPTOM YOU MEASURED	THE UPGRADE THAT FIXES IT	ITS PRICE
Chunks come from the wrong company or the wrong year	metadata filters, agent decides them (built in)	+1 call when used
The right page is retrieved, but ranked fifth, not first	a reranker over the candidate pool	+1 small model
The loop answers too early on a question that needs several hops	a planner, or a sub agent per company, then a merge	+2 to 4 calls
Answers need facts that live outside your documents	corrective web fallback, graded before use	+1 call + the web

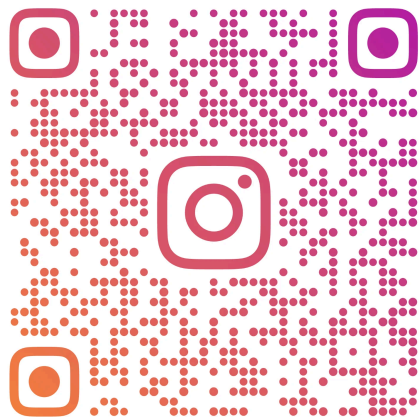
This pipeline has taken **exactly one step** on that ladder: the agent decides the filters. The other three rungs are still waiting for a failure that earns them.

RAG DESIGN · COMPLETE

Thank You

Follow KGP Talkie for new lessons, course updates and live sessions.

Scan to follow



Instagram
@kgptalkie



WhatsApp Channel
KGP Talkie